

Class 10 Science - Chapter 6

Control and Coordination (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Class 7-8 ki yeh cheezein pehle pakki karo, warna Ch6 samajh nahi aayega.

1. Stimulus aur Response (Class 7-8 base)

Stimulus (uddipan) = environment me koi change jis par jeev react kare.

Jaise: garam cheez chhoona, tez roshni, awaaz, smell.

Response (anukriya) = stimulus ke jawab me jeev ka kaam (haath hatana).

Control aur coordination = body ke alag-alag ang milkar (tal-mel se) kaam karein.

2. Cell aur Tissue (Class 8-9 base)

Cell = jeevan ki ikai. Special cell milkar special kaam karte.

Tissue = ek jaise cells ka group jo ek kaam kare.

Nervous tissue = sandesh (message) le jaane wala tissue.

Muscle tissue = sikudkar (contract) movement deta.

3. Gland aur Hormone (idea level)

Gland = woh ang jo koi ras/chemical (secretion) banata.

Exocrine gland: duct (nali) se bahar (saliva, sweat).

Endocrine gland: seedha BLOOD me chemical (hormone) chhodta.

Hormone = chemical messenger jo blood ke through door jaakar control kare.

4. Electric signal vs Chemical signal

Nervous system = TEZ + short-lasting (electrical impulse).

Hormonal system = SLOW + long-lasting (chemical, blood se).

Dono milkar body ko control karte hain.

5. Plant ka idea (Class 7 link)

Plants me brain/nerves nahi hote, par woh stimulus pe react karte (jaise light ki taraf mudna). Yeh hormones (chemical) se hota.

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

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[GREEN = board exam me baar-baar poocha jaata hai]

1. Animals me Nervous System

Control aur coordination 2 system se: NERVOUS system + ENDOCRINE (hormone).

NEURON = nervous system ki structural+functional ikai (sandesh le jaane wali cell).

2. Neuron ki Structure [VERY IMPORTANT]

Neuron ke 3 bhaag:

(1) DENDRITE: information (stimulus) ko receive karta (jhaad jaise sire).

(2) CELL BODY (cyton): info ko aage process/le jaata.

(3) AXON: info ko aage agle neuron tak bhejta (lambi taar jaise).

SYNAPSE = do neuron ke beech ki gap, jahan signal electrical se CHEMICAL ban kar paar hota (neurotransmitter chemical release hota).

Path of impulse: Dendrite -> Cell body -> Axon -> Synapse -> agla neuron.

3. Reflex Action aur Reflex Arc [TOP EXAM AREA]

Reflex action = stimulus par turant, apne-aap (bina soche) hone wali kriya.

Eg: garam cheez chhote hi haath jhatak se hat-na, kaante par pair uthana.

Faayda: brain tak sochne ka time nahi -> bahut tezi se bachav.

REFLEX ARC = woh chhota raasta jisse reflex hota (SPINAL CORD se control):

Receptor (skin) -> Sensory neuron -> Spinal cord -> Motor neuron -> Effector (muscle). Brain ko baad me pata chalta.

Spinal cord reflex ka center hai (brain involve nahi -> isiliye fast).

4. Human Brain [VERY IMPORTANT]

Brain ke 3 main bhaag:

(1) FORE-BRAIN (Cerebrum): sochna, yaad, decision, sensory info, voluntary action. "Sabse bada + smart" hissa.

(2) MID-BRAIN: kuch reflex (eye/pupil) ko control.

(3) HIND-BRAIN:

- CEREBELLUM: body balance + precise/clean movement (cycle chalana).

- MEDULLA (medulla oblongata): involuntary - dhadkan, saans, BP, vomiting, chhk.

- PONS: saans control me madad.

Brain ki suraksha: skull (haddi ka box) + CSF (cerebrospinal fluid - cushion).

Spinal cord ki suraksha: vertebral column (reedh ki haddi).

5. Voluntary, Involuntary, Reflex action

Voluntary action: hamari marzi se (likhna, daudna) - cerebrum control.
 Involuntary action: marzi ke bina (dil dhadakna, saans) - medulla control.
 Reflex action: turant, automatic (garam chhote hat-na) - spinal cord.

6. Coordination in Plants [IMPORTANT]

Plants me nervous system nahi, sirf CHEMICAL (hormone) control. Do tarah:

(A) MOVEMENT WITHOUT GROWTH (immediate):

Touch-me-not (Chhui-mui/Mimosa) ki patti chhoone par band ho jaati - yeh cells me paani ke change se (turant), growth se nahi.

(B) MOVEMENT DUE TO GROWTH = TROPISM (dheere, direction me):

- Phototropism: light ki taraf (tana/stem) ya door (jad) mudna.
- Geotropism: gravity ke hisaab se (jad neeche, tana upar).
- Hydrotropism: paani ki taraf jad mudna.
- Chemotropism: chemical ki taraf (pollen tube ka ovule taraf badhna).

7. Plant Hormones (Phytohormones) [VERY IMPORTANT - RATTA]

Hormone	Kaam
AUXIN	cell ko lamba karta; phototropism (light side kam, doosri side jyada -> stem light ki taraf jhukta)
GIBBERELLIN	stem/growth badhata
CYtokinin	cell division (zyada fruit/seed me)
ABSCISIC ACID	growth ROKTA; patti/stomata band, wilting (stress hormone)

Yaad: "Auxin+Gibberellin+Cytokinin = badhane wale; Absciscic = rokne wala".

8. Human Endocrine System (Hormones) [TOP EXAM AREA - RATTA]

Endocrine gland -> hormone seedha blood me -> target organ par asar.

Gland	Hormone	Kaam
PITUITARY	Growth hormone	body ki growth (kami=dwarf, jyada=gigantism). "Master gland".
THYROID	Thyroxine	metabolism control; IODINE chahiye; kami se GOITRE (gala soojna/ghenga).
PANCREAS	Insulin	blood SUGAR control; kami = DIABETES.
ADRENAL	Adrenaline	"fight or flight" - darr/gusse me heartbeat+BP+saans badhata.
TESTIS (male)	Testosterone	male features (daadhi, awaaz bhari).
OVARY (female)	Oestrogen	female features; menstrual cycle.

Iodised salt kyun? -> Thyroxine ke liye iodine, warna goitre.

FEEDBACK MECHANISM: hormone ki maatra body khud control karti (jaise blood sugar badhe to insulin jyada, sugar normal hote hi insulin kam).

9. Quick Revision - One Liners

- Neuron: dendrite (receive)->cell body->axon->synapse.
- Reflex arc: receptor->sensory->spinal cord->motor->effector.
- Cerebrum (sochna), cerebellum (balance), medulla (saans/dhadkan).
- Plant hormones: auxin/gibberellin/cytokinin (badhao), absciscic (roko).

- Insulin (sugar), thyroxine (iodine/metabolism), adrenaline (fight-flight).

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Reflex arc poora path + spinal cord kyun

Q: Garam tave ko chhote hi haath turant hat jaata. Reflex arc ka pura path likho aur batao brain kyun involve nahi hota.

Solution:

- Path: Receptor (skin) -> Sensory neuron -> Spinal cord -> Motor neuron -> Effector (haath ke muscle).
- Spinal cord turant motor neuron ko signal deta -> haath hat jaata.
- Brain tak signal jaane + sochne me time lagta; jalan se bachne ke liye itna time nahi hota -> isliye spinal cord (fast) handle karta. Brain ko baad me pata chalta.

Example 2 - Phototropism + auxin ka role

Q: Plant ka tana light ki taraf kyun jhukta? Auxin ka role samjhao.

Solution:

- Light padne par auxin hormone shaded (andheri) side me jyada jama hota.
- Jyada auxin -> us side cells jyada lambe -> stem light ki taraf jhuk jaata.
- Yeh PHOTOTROPISM (growth ki wajah se movement) hai.

Example 3 - Nervous vs hormonal coordination

Q: Nervous aur hormonal (endocrine) control me 3 antar likho.

Solution:

- Speed: nervous TEZ (electrical), hormonal SLOW (chemical, blood se).
- Duration: nervous short-lasting, hormonal long-lasting.
- Path: nervous neuron se; hormonal blood se poore body me.

Example 4 - Hormone ka kaam match karna

Q: Insulin, thyroxine, adrenaline ka kaam ek-ek line.

Solution:

- Insulin (pancreas): blood sugar control; kami = diabetes.
- Thyroxine (thyroid): metabolism; iodine chahiye; kami = goitre.
- Adrenaline (adrenal): emergency me heartbeat/BP/saans badhata.

Example 5 (EASIEST) - Neuron ke bhaag

Q: Neuron ke teen main bhaag ke naam.

Solution:

Dendrite, Cell body (cyton), Axon.

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Stimulus aur response ka matlab.
- Q2. Neuron kya hai? Synapse kya hai?
- Q3. Master gland kise kehte hain?
- Q4. Tropism kya hai? 2 example.

MEDIUM (2-3 marks)

- Q5. Reflex action kya hai + 2 example + faayda.
- Q6. Brain ke 3 bhaag + ek-ek kaam.
- Q7. 4 plant hormones aur unke kaam.
- Q8. Nervous aur hormonal control me 3 antar.

HARD (3-5 marks, board favourite)

- Q9. Reflex arc ka pura path + spinal cord kyun (brain nahi).
- Q10. Phototropism + auxin ka role (stem light ki taraf kyun).
- Q11. Human endocrine: 4 gland, hormone aur kaam.
- Q12. Iodised salt kyun zaroori? Thyroxine + goitre samjhao + feedback.

ANSWER HINTS (PART D)

- A1. Stimulus = environment ka change; Response = uska jawab/reaction.
- A2. Neuron = nerve cell (sandesh le jaane wali); Synapse = 2 neuron ki gap.
- A3. Pituitary gland.
- A4. Growth ki wajah se direction me movement; phototropism, geotropism.
- A5. Turant automatic kriya (garam chhote hatna, kaante par pair); fast bachav.
- A6. Cerebrum (sochna), cerebellum (balance), medulla (saans/dhadkan).
- A7. Auxin (cell lamba), gibberellin (growth), cytokinin (division), abscisic acid (rohta).
- A8. Nervous tez+short+neuron; hormonal slow+long+blood.
- A9. Receptor->sensory->spinal cord->motor->effector; fast bachav, brain slow.
- A10. Auxin shaded side jyada -> wahan cell lambe -> stem light ki taraf.
- A11. Pituitary(growth), thyroid(thyroxine), pancreas(insulin), adrenal (adrenaline).
- A12. Thyroxine ke liye iodine; kami se goitre; feedback se hormone level control.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

Generated by Kiro for Krishna1k
